

MagStealth: A Stealth Command Injection Attack on Voice Assistants via Magnetic-Driven Inner Loudspeakers

Tiantian Liu¹, Peng Huang¹, Feng Lin¹, *Senior Member, IEEE*, Tao Ni², and Liang Xiao¹, *Fellow, IEEE*

Abstract—Voice assistants embedded in modern smart devices remain vulnerable to previously unexplored hardware-level attacks. This paper presents MagStealth, a stealthy magnetic injection attack that manipulates built-in loudspeakers to generate malicious commands internally, thereby compromising the associated voice assistant. MagStealth exploits the electromagnetic–acoustic transduction mechanism and validates its feasibility through comprehensive simulations and real-world experiments. To enable practical and covert execution, we design a software-based signal generation framework that introduces adversarial perturbations and develop a compact hardware prototype that can be concealed within power sockets. This design allows MagStealth to bypass physical obstructions, ambient noise, and even speaker verification mechanisms. Extensive evaluations across 25 commercial devices, including smartphones, laptops, tablets, and smart speakers, demonstrate that MagStealth achieves an average injection success rate of 95 percent while maintaining high audio fidelity without acoustic leakage. This work reveals a new class of magnetic attack vectors and highlights the need for hardware-aware defenses in voice-controlled systems.

Index Terms—Out-of-band injection, voice security.

I. INTRODUCTION

LOUDSPEAKERS and microphones are the fundamental components of an audio system, responsible for both audio input and output. These components are integrated with smart voice assistants, providing hands-free interaction to meet human needs for communication and entertainment. Voice assistants, built on loudspeaker-microphone systems, have become ubiquitous in a wide range of electronic devices, such as Apple Siri [1], Google Assistant [2], and Amazon Alexa [3]. Furthermore, these

voice assistants are closely integrated with the Internet of Things (IoT) and vehicles, powering smart homes, intelligent driving, and smart cities.

However, the widespread deployment of voice assistant-enabled devices raises significant concerns regarding privacy and security [4], [5]. Attackers can potentially inject malicious voice commands into the audio system through sound or light modulation, triggering unauthorized actions. For example, researchers have shown that attackers can exploit the non-linearity of microphones [6], [7], [8] to deliver arbitrary commands via modulated ultrasound. Despite the potential of these methods, they come with several limitations: (1) Audible signals crafted using adversarial techniques [9], [10], [11], [12] to confuse voice assistants are often overwhelmed by ambient noise. (2) Ultrasound-based injection attacks suffer from rapid attenuation when passing through materials like plastic and clothing due to their limited penetrability [13]. (3) Some techniques require the assembly of multiple attack devices, which increases the risk of detection [14], [15]. (4) Other methods rely on high-cost setups, such as using the Kepler telescope for laser-based attacks [16], [17]. (5) Most importantly, existing injection attacks [18] fail to address the challenge of bypassing the speaker identification systems built into voice assistants. These limitations highlight the need for a novel command injection attack that can overcome these obstacles and provide deeper insights into the vulnerabilities of audio systems.

In response to these challenges, we propose MagStealth, a novel magnetic injection attack that uses magnetic fields to compromise built-in loudspeakers, bypassing obstacles such as noise, occlusions, and speaker identification systems. To implement MagStealth in practice, we address several key questions: (1) Can loudspeakers be affected by magnetic forces, and if so, how can we manipulate their sound production to generate specific voice commands? To investigate this, we conduct a simulation study that demonstrates how loudspeakers can be influenced by magnetic forces, turning them into a backdoor for magnetic injection attacks. We quantify the impact of external magnetic fields on loudspeakers by modeling the electro-magnetic-acoustic coupling mechanism. By utilizing electromagnetic induction, we show that the voice coil inside the loudspeaker vibrates in response to magnetic fields, producing sound that corresponds to the variations in the field. This demonstrates that a varying magnetic field can manipulate the loudspeaker to emit specific sounds. (2) How can we stealthily bypass

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Tiantian Liu and Liang Xiao are with the Department of Information and Communication Engineering, Xiamen University, Xiamen 361005, China (e-mail: tiantianliu@xmu.edu.cn; lxiao@xmu.edu.cn).

Peng Huang and Feng Lin are with the State Key Laboratory of Blockchain and Data Security, School of Cyber Science and Technology, Zhejiang University, Hangzhou 310000, China, and also with the Hangzhou High-Tech Zone (Binjiang) Institute of Blockchain and Data Security, Zhejiang University, Hangzhou 310007, China (e-mail: penghuang@zju.edu.cn; flin@zju.edu.cn).

Tao Ni is with the Computer, Electrical and Mathematical Sciences and Engineering (CEMSE) Division, King Abdullah University of Science and Technology (KAUST), Thuwal 23955, Saudi Arabia (e-mail: tao.ni@kaust.edu.sa).

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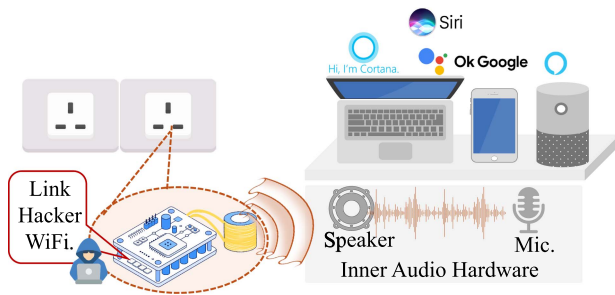


Fig. 1. Attack scenario of MagStealth. The self-designed MagStealth prototype, concealed inside a power socket, emits modulated magnetic fields that induce the nearby device's loudspeaker to produce attacker-crafted audio. The emitted sound is then captured by the device's voice assistant, which subsequently executes the malicious command.

the speaker identification system during an over-the-air attack, without prior knowledge of the target? We design an adversarial signal generation module that crafts imperceptible perturbations to evade speaker verification systems. The module employs a diffusion-based voice conversion decoder to retain the target timbre while optimizing the perturbations against an ensemble of speaker identification models to ensure transferability. Additionally, we incorporate real-world channel distortions to ensure robust over-the-air injection, making the attack both stealthy and effective. (3) How can we launch injection attacks stealthily, without direct intervention, in real-world scenarios? To achieve stealthy magnetic attacks, we design a MagStealth prototype concealed within a modified charging socket. This prototype automatically emits modulated magnetic fields to trigger malicious tasks on nearby victim devices, as shown in Fig. 1. Given the inherent distortion and attenuation of emitted magnetic fields due to limited space, we exploit integrated hardware design to enhance the power amplification module. This improves the intensity of the magnetic fields, extending the attack range. The finely tuned waveform of the signal ensures high-quality and intelligible injections. Notably, the MagStealth prototype operates autonomously through hardware instructions, allowing the attacker to remotely initiate the attack via Bluetooth.

MagStealth introduces a novel threat to electronic devices, expanding the attack surface and providing new insights into hardware security design.

In summary, our contributions are as follows:

- MagStealth introduces an aggressive magnetic threat to voice assistants via a loudspeaker-based backdoor, bypassing existing speaker identification systems. We explore the feasibility of magnetic injection attacks and provide detailed theoretical guidance through simulations and experiments, modeling the relationship between external magnetic fields and loudspeakers.
- We design a prototype to implement stealthy offline magnetic attacks, leveraging an integrated system design methodology to create low-cost and compact attack setups.
- We implement and evaluate the MagStealth prototype on 25 voice-enabled devices, including smartphones, tablets, laptops, and smart speakers. Experimental results show that

MagStealth achieves an average success rate of 95%, and is resilient to noise, occlusions, and electromagnetic leakage.

II. BACKGROUND

This section presents the theoretical foundations of magnetic fields, covering electromagnetism and electromagnetic induction, and examines the resulting vulnerabilities of loudspeakers that can be exploited in magnetic-based attacks.

A. Introduction to Magnetic Fields

A magnetic field is a physical field generated by moving electric charges, electric currents, and magnetic materials, capable of exerting forces on other magnetic objects within its influence [19]. Magnetic fields propagate through various media, including air, solid materials, and even vacuum, without requiring a physical medium.

Electromagnets: An electromagnet is typically constructed by winding a conductive wire into a solenoid around a magnetic core. When an electric current passes through the coil, it induces a concentrated magnetic field, the strength of which is proportional to the current magnitude [20]. A steady current generates a static magnetic field, whereas a time-varying current produces a dynamic magnetic field. The field intensity diminishes with distance from the source. Due to their low cost, design flexibility, and precise controllability, electromagnets are widely employed in electromechanical systems such as transformers, electric motors, generators, and loudspeakers [21]. In this work, we adopt an electromagnet as the primary source of magnetic fields for injection attacks.

Electromagnetic induction: Electromagnetic induction refers to the generation of an electromotive force (EMF) across an electrical conductor exposed to a changing magnetic field [22]. This principle reveals the fundamental coupling between magnetism and electricity, manifesting in two primary forms: (1) when a conductor moves relative to a static magnetic field, or (2) when a time-varying magnetic field surrounds a stationary conductor. In both cases, an EMF is induced, and if the conductor forms a closed circuit, an electric current flows. The magnitude of the induced EMF is proportional to the rate of change of magnetic flux through the conductor. In essence, electromagnetic induction enables the conversion of magnetic energy into electrical energy within a closed circuit.

B. Layout of Audio Systems in Smart Devices

Magnetic principles in loudspeakers: A loudspeaker functions as an electroacoustic transducer that converts alternating electrical signals into corresponding sound waves [23]. As illustrated in Fig. 2(a), a typical dynamic loudspeaker comprises a diaphragm, a voice coil, and a permanent magnet. The permanent magnet establishes a static magnetic field between its annular and central poles. The voice coil, wound around the central pole and suspended within this field, acts as an electromagnet when an alternating current passes through it. The resulting varying magnetic interaction causes the coil to oscillate axially, driving the attached diaphragm to vibrate. These vibrations produce

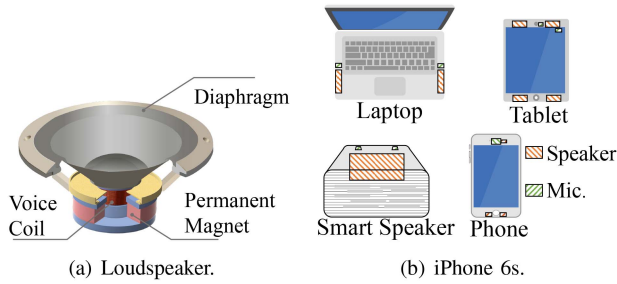


Fig. 2. An illustration of the loudspeaker's internal structure and the spatial arrangement of loudspeakers and microphones in smart devices.

pressure waves in the surrounding air, thereby generating sound. This inherent magnetic sensitivity makes loudspeakers susceptible to external magnetic fields, allowing potential manipulation of sound output through carefully crafted magnetic signals.

Audio systems in commercial smart devices: In modern consumer electronics such as smartphones, smart speakers, and laptops, loudspeakers are commonly integrated with microelectromechanical system (MEMS) microphones to form a complete audio subsystem [24]. This combination supports both audio playback and capture, enabling voice-controlled functionalities including speech recognition, command execution, and system feedback. On typical device circuit boards, the loudspeaker and microphone are placed only a few millimeters apart. This proximity stems from design constraints: (1) cost and size considerations often lead to shared audio codecs and sound cards between the two components, and (2) loudspeakers and microphones may also share a common acoustic port or sound channel [25]. Fig. 2(b) illustrates this close arrangement in various devices such as laptops and smart speakers. For instance, in smartphones, the front and bottom microphones are frequently located adjacent to loudspeaker units. Due to this physical closeness, sound emitted by the loudspeaker is inevitably captured by nearby microphones. This architectural characteristic motivates our approach of exploiting the loudspeaker as an indirect injection channel to deliver malicious voice commands to the microphone.

III. THREAT MODEL

In this threat model, an adversary aims to inject unauthorized voice commands into voice-controlled devices by leveraging externally generated magnetic fields. The objective is to compromise such devices, typically equipped with integrated microphones and loudspeakers, and induce them to perform malicious actions that may lead to privacy breaches or financial loss.

Victim device: The targeted devices are widely used consumer electronics that incorporate both loudspeakers and microphones, such as smartphones, tablets, laptops, and smart speakers. These devices generally run popular voice assistant services (e.g., Siri, Google Assistant, Alexa). The attacker is assumed to have no prior physical or software access to the device. That is, they cannot modify system settings, install malicious applications, or tamper with the device hardware. During the attack, the target

device is assumed to be resting on a typical indoor surface, such as a desk, table, or charging platform.

Attacker's capability: The attacker is assumed to have the capability to generate controlled magnetic signals and craft adversarial audio waveforms, but does not possess direct physical access to the internal hardware of the victim device nor the ability to install malicious software. The attacker possesses the capability to design and assemble a portable magnetic-field-emitting device capable of generating carefully modulated alternating magnetic fields. This device can produce magnetic signals that penetrate the casing of the target device and induce controlled vibrations in its built-in loudspeaker, effectively transforming it into an unintended sound source. By precisely controlling the magnetic field waveform, the attacker can cause the loudspeaker to reproduce valid voice commands, thereby misleading the voice recognition system. The attacker is also assumed to have access to publicly available voice recordings, such as those captured in public spaces or uploaded by users on social media, enabling them to obtain 1-2 voice samples that are sufficient for protecting the victim's voice timbre. Once deployed, the attack device operates autonomously without requiring the attacker's physical presence or manual intervention, enhancing both scalability and stealth.

Attack scenarios: The attack device is designed to be compact and concealable, allowing it to be embedded inside common household or public objects, e.g., within a wall power socket, a charging dock, or a piece of furniture. Disguised as an ordinary charging socket, it can be deployed in locations such as cafes, libraries, offices, or residential spaces where users routinely place their devices for charging. When a victim device is positioned near the compromised socket, the hidden attack hardware emits magnetic fields modulated with malicious voice commands. A typical attack sequence may begin with a command to lower the device's media volume, reducing the likelihood of auditory detection. Subsequent commands could instruct the device to make unauthorized purchases, enable Bluetooth or Wi-Fi for data exfiltration, open vulnerable applications, or initiate other actions that benefit the attacker.

IV. SIMULATION STUDY OF MAGNETIC ATTACK

To demonstrate the feasibility of manipulating loudspeaker acoustics using externally applied magnetic fields, we conduct a set of simulation experiments in COMSOL Multiphysics [26]. In these simulations, a typical dynamic loudspeaker is placed within a uniformly distributed magnetic field oriented vertically so that the field lines pass through the diaphragm from top to bottom. By adjusting the frequency and amplitude of the applied field, we generate either a static magnetic field or various time-varying magnetic fields to perturb the loudspeaker. The simulation parameters are summarized in Table I. For each configuration, we estimate both the frequency and the sound pressure level (SPL) of the acoustic signals emitted by the loudspeaker under different external magnetic excitations.

The resulting acoustic fields are illustrated in Figs. 3 and 4. Simulation results indicate that under a time-varying magnetic

TABLE I
SIMULATION PARAMETERS OF THE LOUDSPEAKER IN A MAGNETIC FIELD

Type	Element	Parameter	Value
Loudspeaker	Magnet	Remanent flux density	0.4 T
	Voice coil	Number of turns	100
	Voice coil	Relative permeability	1 H/m
	Diaphragm	Young's modulus	2 GPa
Magnetic field	Intensity	Frequency	0-8 kHz
Magnetic field	Intensity	Amplitude	0-0.04 T

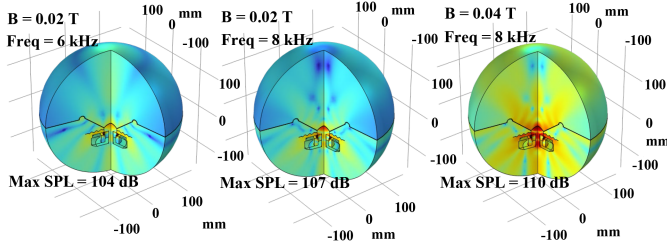


Fig. 3. Partial simulation results of estimated sound from the loudspeaker in different frequencies and amplitudes of magnetic fields.

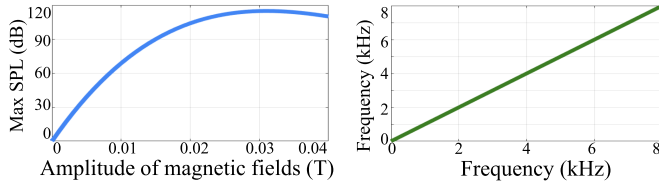


Fig. 4. Max SPL and frequency of sound emitted from loudspeaker under different amplitude and frequency of magnetic fields.

field with applied frequencies of 6 kHz and 8 kHz, the loudspeaker emits sound at corresponding frequencies of 6 kHz and 8 kHz, respectively. To quantify the emitted sound strength, we compute the maximum SPL within the simulated sound field distribution. As the magnetic field amplitude increases from 0 T to 0.02 T, the maximum SPL increases from 0 dB to 104 dB, indicating a positive correlation between the magnetic field amplitude and the resulting acoustic output. In contrast, the loudspeaker emits no sound when subjected to a static magnetic field. These observations confirm that only a dynamic magnetic field can induce audible vibrations in the loudspeaker diaphragm. Based on these simulation results, we adopt a dynamic magnetic field as the attack signal for subsequent experiments.

V. ELECTRO-MAGNETIC-ACOUSTIC COUPLING MODEL

To generate a controlled variation of the magnetic field that induces specific acoustic emissions from a loudspeaker, we establish a mathematical model to describe the coupling between the magnetic field and the loudspeaker system. The loudspeaker is modeled as a simple electromechanical system, with key components consisting of the voice coil and permanent magnet. For the purpose of analysis, the internal loudspeaker system is simplified as shown in Fig. 5. The strength of the magnetic field B at a point r is directly influenced by the current I flowing

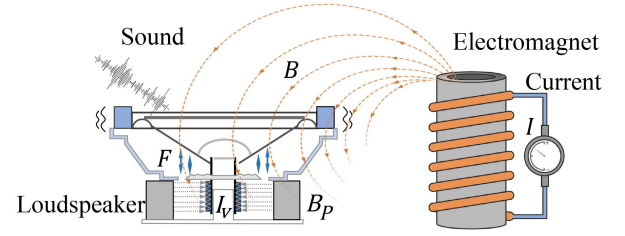


Fig. 5. The mathematical model: A varying magnetic field generated by an electromagnet induces current in the voice coil of the loudspeaker, resulting in sound generation.

through the electromagnet, as described by the Biot-Savart law:

$$B(r) = N_M \frac{\mu I}{4\pi} \int_C \frac{dl \times (r - l)}{|r - l|^3}, \quad (1)$$

where dl represents a differential vector element along the current-carrying path C , l is the position vector of the current element, and $r - l$ is the displacement from the current element to the field point r . N_M denotes the number of turns in the electromagnet's coil, and μ is the permeability of the material. This equation describes the magnetic field distribution generated by the electromagnet. As the magnetic field $B(r)$ varies over time, an electromotive force (EMF) is induced in the voice coil due to electromagnetic induction. According to Faraday's law of induction [27], the induced EMF E in the voice coil is given by:

$$E = N_V \frac{d\Phi}{dt} = N_V N_M \frac{\mu A}{4\pi} \int_C \frac{dl \times (r - l)}{|r - l|^3} \cdot \frac{dI}{dt}, \quad (2)$$

where N_V is the number of turns in the voice coil, and A is the cross-sectional area of the coil. The flux Φ through the coil is linked to the change in magnetic field. Consequently, the induced current I_V in the voice coil can be expressed as:

$$I_V = \frac{E}{R} = \frac{1}{R} N_V N_M \frac{\mu A}{4\pi} \int_C \frac{dl \times (r - l)}{|r - l|^3} \cdot \frac{dI}{dt}, \quad (3)$$

where R is the resistance of the voice coil. The induced current in the voice coil generates a force within the loudspeaker system due to the interaction of this current with the permanent magnetic field. This force, often referred to as the Ampere force, can be calculated as:

$$F = B_P I_V L = \frac{B_P L}{R} N_V N_M \frac{\mu A}{4\pi} \int_C \frac{dl \times (r - l)}{|r - l|^3} \cdot \frac{dI}{dt}, \quad (4)$$

where L is the length of the voice coil, and B_P is the strength of the permanent magnetic field in the loudspeaker. Note that this Ampere force is generated by the permanent magnet, not the external electromagnet, as the direction of the external magnetic field would be perpendicular to the motion of the diaphragm of loudspeakers, making it ineffective in generating a driving force. According to (4), the induced force is directly proportional to the time derivative of the driving current, the permanent magnetic field strength, and the effective length of the voice coil. In practical magnetic injection scenarios, the most influential controllable factor is the rate of change of the injected current, which determines the magnitude of the induced current in the

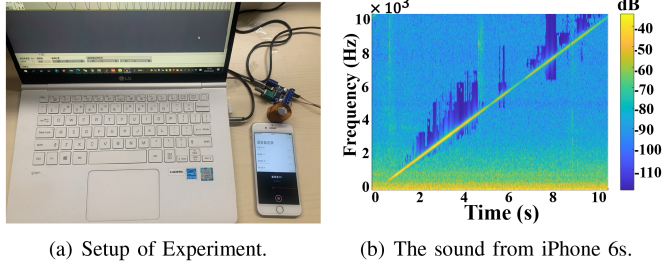


Fig. 6. The setup of injection experiment and the injected result from iPhone 6s.

voice coil and therefore the resulting acoustic vibration. For typical smartphone loudspeakers, the permanent magnetic field strength B_P ranges from 0.8T to 1.2T for neodymium magnets. The voice coil length L typically varies between 1.5m and 3.0m, depending on the number of turns and coil diameter, while the voice coil resistance R commonly falls within the range of 4 Ω to 8 Ω for consumer electronics. These values yield induced forces on the order of millinewtons, sufficient to generate audible sound pressure levels above 70 dB at close proximity.

The Ampere force defined in (4) is directly proportional to the time derivative of the external current, governing both the amplitude and frequency of the resulting force. This force drives the vibration of the voice coil and the attached diaphragm, thereby generating acoustic pressure waves. The sound pressure output $p(t)$ is thus proportional to the applied force and, consequently, to the current derivative:

$$p(t) \propto F(t) \propto \frac{di(t)}{dt}. \quad (5)$$

This fundamental relation reveals that the loudspeaker's acoustic output is directly shaped by the time-varying profile of the external current. As a result, by strategically modulating the current's derivative, arbitrary audio waveforms can be synthetically induced in the loudspeaker via electromagnetic coupling. This principle forms the theoretical foundation for electromagnetic acoustic injection, wherein tailored current signals are designed to reproduce specific acoustic commands through magnetic transduction.

Experimental validation: To validate the effectiveness of this model in real-world conditions, we conduct an experiment to test whether magnetic fields can induce acoustic signals in smartphones. The experimental setup is shown in Fig. 6. In this experiment, a TPA3116D2 amplifier board [28] is connected to a laptop's sound card, driving the electromagnet to produce a variable magnetic field. The magnetic field is modulated in consistent with a 0-10 kHz chirp signal, which is used to interfere with an inner loudspeaker. The iPhone 6s is used as the target device for the magnetic injection attack. The results for injected signals into the iPhone 6s are shown in Fig. 6(b). The captured audio reveals the presence of nonlinear distortion and background noise in the magnetically induced signal. Although the induced audio waveform generally follows the variations in the magnetic field, the transduction process introduces non-negligible transmission distortion. These findings highlight the

need for sophisticated signal design that accounts for channel distortion and incorporates robustness against such impairments to ensure reliable magnetic injection.

VI. MAGSTEALTH DESIGN

MagStealth is an attack framework that integrates software-based adversarial signal generation with a compact, concealable hardware platform to enable stealthy magnetic command injection, as shown in Fig. 7. The system operates through a coordinated pipeline: first, a diffusion-based adversarial audio generator crafts imperceptibly perturbed voice commands that evade speaker identification models while preserving target timbre. These signals are then modulated and amplified via a custom power-amplification circuit, which drives a precisely optimized electromagnetic transducer to emit time-varying magnetic fields. The hardware prototype, miniaturized into a form factor concealable within common objects such as charging sockets, delivers these fields to the target device, where they induce the built-in loudspeaker to internally reproduce malicious audio.

A. High-Coverttness Signal Generation

To ensure robust magnetic injection in practical scenarios, this module is designed to generate adversarial signals that remain effective against both speaker identification systems and the distortions inherent in over-the-air transmission.

1) *Adversarial Perturbation Generation:* We build the adversarial perturbation generator on top of a conditional diffusion voice-conversion decoder inspired by one-shot diffusion probabilistic modeling, whereby the adversarial audio only entailing one piece of target utterances. Firstly, the pretrained transformer-based encoder same as used in Grad-TTS [29] produces linguistic representations from the source waveform that contains malicious commands meant to be injected. The target speaker embedding is extracted from a pre-collected utterance from a speaker encoder. The high-level linguistic representation and speaker embedding are concatenated as a latent representation that is fed into the following decoder consisting of the U-Net. The decoder reconstructs the adversarial mel-spectrogram preserving the timbre of target speakers. In reverse diffusion of decoder, the adversarial constraint δ is added to generated mel-spectrogram per timestep. Let $\hat{X}_t$ denote the denoised spectrogram at timestep t in diffusion process. At each step we query a speaker classifier $f(\cdot)$ on the current state. If the prediction does not match the desired target label y' , we compute a correction δ by solving a constrained optimization problem:

$$\min_{\delta} \mathcal{L}_{CE} \left(f(\hat{X}_t + \delta), y' \right) \quad \text{s.t. } \|\delta\| \leq \varepsilon, \quad (6)$$

and then update the state as $\hat{X}_t^{\text{adv}} = \hat{X}_t + \delta$. Here, $\mathcal{L}_{CE}$ denotes the cross-entropy loss, which enforces the adversarial examples to drive the speaker identification model toward predicting the target speaker label. The hyperparameter ε constrains the perturbation amplitude to ensure high stealthiness. In our experiments, the perturbation constraint parameter ε is empirically set within the range [0.01, 0.05], which ensures that the adversarial perturbations remain imperceptible while effectively influencing

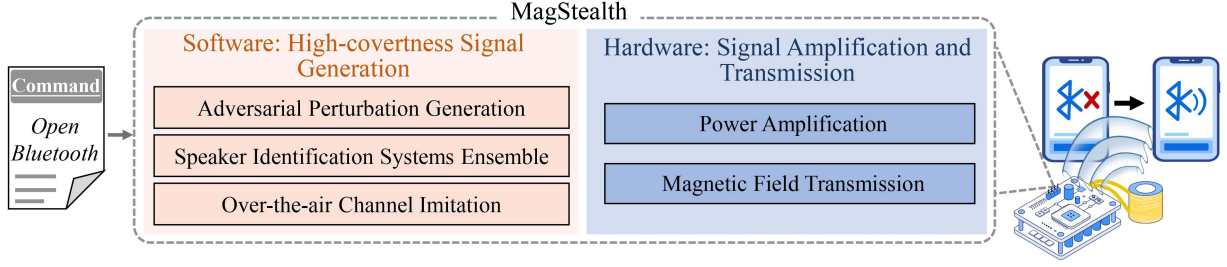


Fig. 7. The system overview of MagStealth. The system integrates dedicated software and hardware modules that operate in coordination to deliver an effective magnetic injection attack.

speaker identification models. The whole training objective of adversarial perturbation generator combines a reconstruction term with the adversarial constraint:

$$\mathcal{L}_{total} = \|\hat{X}_t^{adv} - X_t^{ref}\|_2 + \lambda_{adv}(t) \mathcal{L}_{CE}(f(\hat{X}_t^{adv}), y'), \quad (7)$$

where X_t^{ref} represents the ground truth spectrogram of source speech. The reconstruction loss dominates at early noisy steps to protect timbre characteristics, while $\lambda_{adv}(t)$ gradually increases at later steps to inject adversarial guidance into the fine details of the spectrogram. This scheduling strategy ensures that the perturbations are both effective in misleading the victim speaker identification model and perceptually concealed.

2) *Speaker Identification Systems Ensemble*: To achieve strong transferability across diverse speaker identification systems, the adversarial perturbation should be designed to capture universal features of the target speaker. A straightforward approach is to ensemble multiple speaker identification systems by adaptively reweighting their respective losses. Formally, the perturbations are optimized against an ensemble $\{f_i\}_{i=1}^N$ composed of heterogeneous architectures (e.g., ECAPA-TDNN [30], X-Vector [31], etc.). The ensemble loss is accordingly defined as: $\mathcal{L}_{ens} = \sum_{i=1}^N w_i \cdot \mathcal{L}_{SID}(f_i(\hat{X}_t^{adv}), y')$, where N is the number of surrogate models, and w_i denotes the ensemble weight for model i .

To prevent any single surrogate model from dominating the ensemble, the weights w_i are dynamically adjusted throughout the optimization process. During the iteration of adversarial sample generation, each model loss $\mathcal{L}_{SID}$ is standardized by using its running mean μ_i and variance σ_i . Concretely, let $\tilde{L}_i = (\mathcal{L}_{SID,i} - \mu_i)/(\sigma_i + \kappa)$ be the normalized loss for model i , where $\kappa > 0$ is a small smoothing constant included for numerical stability. The adaptive weights are then updated via a softmax function:

$$w_i = \frac{\exp(\gamma \cdot \tilde{L}_i)}{\sum_{j=1}^N \exp(\gamma \cdot \tilde{L}_j)}, \quad (8)$$

where γ is a temperature parameter that controls the sharpness of weighting. Through the ensemble strategy with adaptive weighting, the generated adversarial examples are more likely to generalize across unseen speaker identification systems, thereby enhancing the effectiveness of opaque-box injection attacks.

3) *Over-the-Air Channel Imitation*: To facilitate practical over-the-air magnetic injections, we imitate the propagation

channel and distortions during the generation of adversarial samples. Given the noise interference on the emitted magnetic signals, this system adds random noise n (e.g., white Gaussian noise) into the perturbation to model ambient noise in the physical world, represented as $\hat{X}_t^{adv} + n$. The transmission channels of MagStealth span both magnetic channel and audio channel. Thus, inspired by previous work on audible channel imitations, we integrate the room impulse response h_{aud} to emulate audible channel distortions. For magnetic channel, according to the electromagnetic radiation mechanism, the response h_{mag} can be described by the following function [32]:

$$h_{mag} = \frac{1}{\alpha t} e^{-\beta t}, \quad (9)$$

where α reflects the distance-dependent decay, and β is the exponential decay rate. Additionally, a band-pass filter $\mathcal{F}$ is applied to the adversarial sample to replicate the effects of received audio in victim microphones. After thoroughly considering the various influencing factors, the adversarial current optimization problem is formulated as:

$$\mathcal{L}_{total} = \|\hat{X}_t^{adv} - X_t^{ref}\|_2 + \lambda_{adv}(t) \mathcal{L}_{ens} \left\{ \mathcal{F} \left((h_{aud} + h_{mag}) \otimes \hat{X}_t^{adv} \right) + n, y' \right\}. \quad (10)$$

Finally, MagStealth can generate adversarial samples aiming at spoofing speaker identification systems. For ease of deployment, the samples that attackers want to inject are pre-stored in an SD card of the system. MagStealth uses a hardware decoding circuit equipped with a digital to analog converter (DAC) to decode digital files of adversarial samples, producing aggressive currents consistent with the expected perturbed audio.

B. Signal Amplification and Transmission

To enable effective magnetic injection, the generated adversarial audio signal must undergo precise power amplification and transmission. This stage ensures that the signal retains its adversarial properties while being delivered efficiently through an electromagnetic channel. The following sections detail the design of our power amplification and the optimization of the electromagnetic transducer for robust and covert signal emission.

1) *Power Amplification*: The audio band is particularly susceptible to distortion under conventional analog amplification. To mitigate this issue, we employ pulse-width modulation

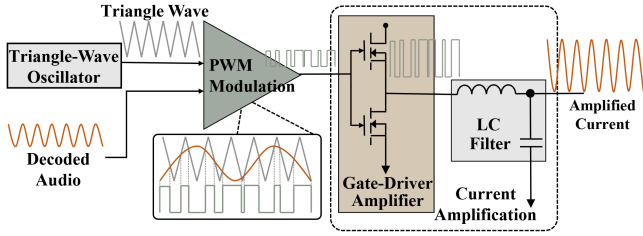


Fig. 8. The signal processing flow of Power Amplification.

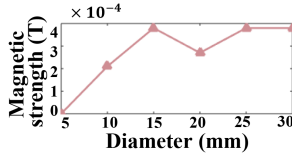


Fig. 9. Measured magnetic strength under different diameters.

(PWM), which encodes the decoded audio signal using a rectangular pulse train whose duty cycle is proportional to the instantaneous amplitude of the audio waveform. Because rectangular pulses have a broad spectrum, PWM preserves the full 20–20,000 Hz audio bandwidth and provides strong robustness against amplitude noise due to its frequency-modulated property. For fast and stable operation, integrated analog circuits are used to implement the PWM logic.

As shown in Fig. 8, a triangle-wave oscillator generates a periodic carrier $c(t)$, which is compared against the decoded audio $a(t)$ using a high-speed comparator. The comparator outputs a binary voltage signal $\delta(t)$ defined by

$$\delta(t) = \text{sgn}(a(t) - c(t)) = \begin{cases} 1, & a(t) > c(t), \\ 0, & a(t) \leq c(t). \end{cases} \quad (11)$$

Whenever the audio amplitude exceeds the instantaneous value of the triangular carrier, the comparator outputs its maximum voltage. Otherwise, it outputs its minimum voltage. The resulting PWM stream is therefore a sequence of pulses whose duty cycle encodes the original audio amplitude.

The PWM signal, although robustly modulated, initially carries low power. The second stage amplifies it into a high-current drive signal while maintaining its waveform fidelity. Because PWM signals feature fast edges and high switching frequencies, the amplifier must support rapid voltage transitions between positive and negative saturation. We employ a gate-driver amplifier that outputs only rail-to-rail voltage levels, functioning effectively as a high-speed power switch. To recover the underlying audio waveform after amplification, we first theoretically analyze the signal in the frequency domain by expanding $\delta(t)$ into its Fourier series representation. With carrier frequency f_c and period $T_c = 1/f_c$, the PWM sequence can be expressed as:

$$\delta(t) = \sum_{k=-\infty}^{+\infty} \tau_k(a(t)) e^{j2\pi k f_c t}, \quad (12)$$

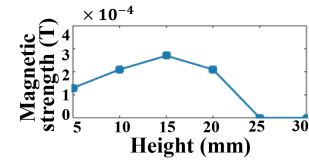


Fig. 10. Measured magnetic strength under different heights.

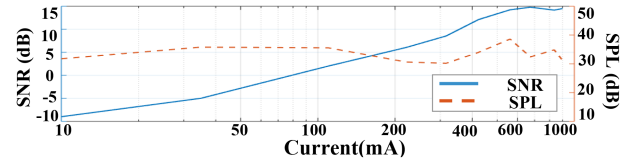


Fig. 11. The estimated SNR of injected signals and SPL around the target. when currents of different strengths are loaded on the electromagnet.

where the Fourier coefficients $\tau_k(a(t))$ are

$$\begin{aligned} \tau_k(a(t)) &= \frac{1}{T_c} \int_{\frac{T_c}{2} - \frac{a(t)T_c}{2}}^{\frac{T_c}{2} + \frac{a(t)T_c}{2}} e^{-j2\pi k f_c t} dt \\ &= \frac{1}{\pi k} \sin(\pi k a(t)) e^{-j\pi k}. \end{aligned} \quad (13)$$

Substituting (13) into (12) yields:

$$\delta(t) = a(t) + \sum_{k=1}^{\infty} \frac{2(-1)^k}{\pi k} \sin(\pi k a(t)) \cos(2\pi k f_c t). \quad (14)$$

Because the desired audio $a(t)$ occupies a bandwidth far below the carrier frequency f_c , it appears only in the low-frequency component of $\delta(t)$. Thus, a simple low-pass filter, implemented with an inductor-capacitor (LC) network, effectively suppresses high-frequency PWM harmonics and extracts the clean analog audio current for subsequent transmission.

2) Magnetic Field Transmission: After signal amplification, the amplified current is applied to a magnetic transducer that continuously emits time-varying magnetic fields. The emitted field strength depends not only on the drive current but also on key physical properties of the transducer. To achieve strong and spatially extended magnetic radiation, we design a customized cylindrical core geometry optimized across four parameters: core diameter, core height, number of coil turns, and driving current.

Diameter of Core: The diameter of the ferromagnetic core determines the magnetic flux density both inside and outside the magnetic structure. We fabricate six soft-iron cylindrical cores with diameters from 5 mm to 30 mm, each wrapped with 20 turns of copper wire and driven with 10 mA current. Measurements in Fig. 9 using a teslameter indicate that the 15 mm core yields the strongest field and is thus selected.

Height of Core: Core height of electromagnet also influences the emitted magnetic flux. We fabricate six cores of height 5 mm–30 mm, all with 10 mm diameter and the same 20-turn winding. Measurements in Fig. 10 show that a 15 mm height maximizes the emitted field.

Number of Coil Turns: The magnetic field strength is proportional to the number of coil turns. To achieve sufficient field intensity and attack range, the electromagnet is wound with 400 turns.

Driving Current: The final magnetic output is proportional to the amplified current. The current must be strong enough to induce detectable internal audio yet low enough to prevent audible leakage outside the device. Through controlled experiments on an iPhone 6 s, sweeping the injected current while measuring the resulting SNR and ambient SPL, we find in Fig. 11 that currents exceeding 40 mA achieve the threshold SNR (> -5 dB) required for voice-activity detection [33], while maintaining an SPL in the 30–40 dB range, comparable to quiet indoor ambient noise.

These design choices together establish a strong and covert magnetic transmission path capable of reliably delivering injected signals into target devices.

VII. HARDWARE OPTIMIZATION AND IMPLEMENTATION

The initial MagStealth prototype is implemented on a large development board measuring approximately $10\text{ cm} \times 10\text{ cm}$. This system employs a modular hardware architecture consisting of an STM32 microcontroller, which serves as the central controller for coordinating subsystems and managing data flow. Audio decoding is performed by a VS1003 dedicated audio decoder chip [34] that receives digital audio files from external storage via a serial peripheral interface and converts them into analog signals. These signals are amplified by a TDA8932 class-D power amplifier [35], which uses pulse-width modulation to drive a custom electromagnet and generate the required time-varying magnetic field. The system includes multiple external storage interfaces, such as USB ports and an SD card slot, along with a liquid crystal display for monitoring operational status. The signal processing chain involves retrieving digital audio files from external storage, decoding them with the VS1003, filtering the output to suppress high-frequency pulse-width modulation components, amplifying the signal with the TDA8932, and delivering it to the electromagnet. Although this modular design provides comprehensive functionality and debugging capabilities, the multiple discrete components, external interfaces, and visible display result in a large and conspicuous device that is unsuitable for covert deployment.

To achieve the stealth requirements for practical attack scenarios, the hardware is optimized into a compact miniature device measuring $6\text{ cm} \times 3\text{ cm} \times 1.5\text{ cm}$. This optimization consolidates the previously distributed decoding functions, which are spread across the STM32 microcontroller, VS1003 chip, and multiple external storage interfaces, into a single BK3266 system-on-chip [36]. The BK3266 integrates a hardware audio decoder, Bluetooth 4.2 module, and sufficient internal memory to enable direct reading and decoding of pre-stored adversarial audio files from an embedded SD card via its general-purpose input/output pins. The decoded analog signal is transmitted to the amplification circuitry on a double-sided printed circuit board, thereby eliminating the need for separate control components and external interfaces. The amplification circuitry retains the

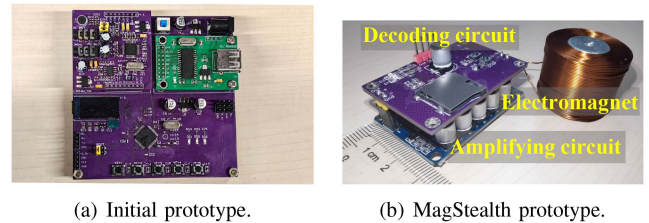


Fig. 12. The implementation of MagStealth.

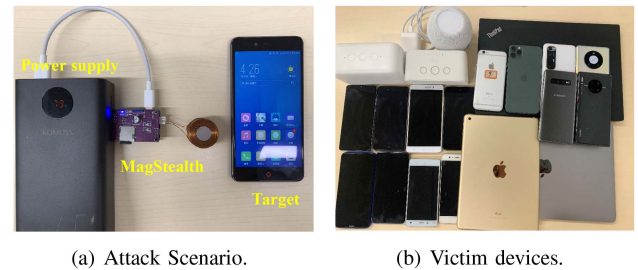


Fig. 13. The evaluation of setup.

TDA8932B class-D amplifier as the primary power amplification element. To maximize output power within the limited space, the dual output channels are connected in parallel, which allows their current signals to superimpose and increase drive capability. Five parallel polarized capacitors are connected to the amplifier's power input for ripple suppression, and two $10\text{ }\mu\text{H}$ inductors, each paired with a film capacitor, form a low-pass filter to remove high-frequency pulse-width modulation carrier components. The amplified current is then supplied to the custom electromagnet. Power distribution is managed by a CH224 K USB Type-C power delivery controller [37].

By replacing the distributed multi-component architecture with a highly integrated system-on-chip, eliminating all user interface elements such as the display and external storage interfaces, and implementing a compact double-sided printed circuit board layout, the system is transformed from a large, visibly operational development platform into a discreet, autonomous miniature device. This optimized configuration maintains the essential signal processing chain required for effective magnetic injection while removing physical and operational indicators that could reveal the device's presence. Consequently, the resulting prototype achieves a substantial reduction in size and enhanced covertness, making it suitable for deployment in realistic attack scenarios where device concealment is essential.

VIII. EVALUATION

A. Experimental Settings

Victim Setup: The evaluation of MagStealth is conducted on a range of commercial electronic devices, including smartphones, smart speakers, laptops, and smart tablets, as shown in Fig. 13. These victim devices, which incorporate both loudspeakers and microphones, are produced by major manufacturers such as Apple, Google, Samsung, Huawei, and Xiaomi. For devices

TABLE II
LIST OF MALICIOUS VOICE COMMANDS PRE-STORED IN MAGSTEALTH,
WAITING TO BE INJECTED INTO THE DEVICE

The content of voice commands	
Open the door	Call my boss
Open Instagram	Set an alarm
Turn on Bluetooth	Purchase an iPad Pro
What time is it	What's forty four plus ninety three

equipped with voice assistants, such as Siri, the voice assistants are fully enrolled by their respective owners. Additionally, four state-of-the-art speaker identification models are employed in the evaluation: X-Vector [38], D-TDNN [39], ECAPA-TDNN (ECAPA) [40], and AutoSpeech. These models are trained using the additive angular margin softmax loss function.

Attacking Setup: The MagStealth prototype, as illustrated in Fig. 12(b), is used to inject malicious commands into a target victim device. Audio waveforms representing malicious voice commands are generated by using high-covertiness signal generation module. These commands, stored in MP3 format with a sampling rate of 48 kHz, are pre-recorded on an SD card within the decoding module. The text of the malicious commands is provided in Table II. During magnetic injection attacks, each command is preceded by the appropriate wake word for the target device, such as “Hey Siri.” The prototype is powered through a USB Type-C interface, with a power supply of 12 V and 1 A.

Through systematic frequency sweeps from 100 Hz to 20 kHz across tested devices, we find that the effective bandwidth of magnetic-to-acoustic transduction is primarily limited by the loudspeaker’s electromechanical response rather than the microphone or amplification stage. Smartphone loudspeakers typically operate effectively between 300 Hz and 8 kHz, with attenuation at low and high frequencies due to diaphragm mass and voice coil inductance. Smart speakers extend to lower frequencies around 80 Hz but exhibit stronger high-frequency attenuation above 5 kHz. In contrast, microphones maintain a relatively flat response over the audible range and do not constrain performance within the loudspeaker’s operating band. The amplification stage in MagStealth provides a near-flat response across the audio spectrum, ensuring minimal distortion before magnetic emission. As a result, command intelligibility depends mainly on the loudspeaker passband. Based on this observation, we design adversarial commands to concentrate energy in the 300 Hz to 4 kHz range to ensure robustness and intelligibility across all tested devices.

Metrics: The effectiveness of the magnetic injection attacks is evaluated using the injection success rate. This metric is defined as the ratio N_r/N , where N_r denotes the number of injection attempts that are correctly recognized by both the speaker identification system and the speech recognition system of the target device, and N represents the total number of injection attempts. To assess the stealthiness of the adversarial voices generated by MagStealth, two objective metrics are employed: the SPL and the perceptual evaluation of speech quality (PESQ) [41]. During the injection process, the maximum SPL measured among the

victim devices and the attacking devices is recorded as the relevant metric. The PESQ score provides an objective evaluation of the quality of the audio signal captured by the microphone of the victim device. PESQ scores range from -0.5 to 4.5 , with higher scores corresponding to better perceived audio quality. Lower SPL values combined with higher PESQ scores indicate greater stealthiness of the adversarial audio.

For smart speakers (HomePod mini, MI Smart Speaker, TmallGenie), PESQ scores are marked as “N/A” because these devices do not provide accessible audio recording interfaces or diagnostic ports that would enable direct capture of the microphone-received signal. Without a ground-truth reference recording synchronized with the injected command, PESQ computation is not feasible. Therefore, we report only injection success rate and ambient SPL for these devices.

B. Overall Performance

We comprehensively evaluate the overall effectiveness and stealthiness of MagStealth across four mainstream smart device categories: smartphones, smart speakers, tablets, and laptops. Specifically, our evaluation targets a total of 16 smartphones, two tablets, four laptops, and three smart speakers. The MagStealth prototype is positioned at a default distance of 25 mm from each target device. For each device, we conduct magnetic injection attacks 100 times per command and measure the injection success rate, SPL, and PESQ. Detailed results are summarized in Table III. The experimental results demonstrate that MagStealth achieves an average injection success rate of 95.01% across all tested devices, independent of the underlying speaker identification systems. The SPL during injection remains consistently around 35 dB, which corresponds to quiet environmental noise, confirming the high stealthiness of the attack as it is unlikely to alert nearby individuals. When tested against both commercial voice assistants and SOTA academic speaker identification systems, MagStealth demonstrates outstanding performance, effectively bypassing these systems while maintaining high PESQ scores. This indicates that the adversarial voice commands generated by MagStealth retain excellent speech quality despite their aggressiveness.

The observed performance variation across device categories can be attributed to several hardware and structural factors. Smart speakers exhibit lower success rates compared to smartphones. This is primarily due to their larger loudspeaker drivers, which typically measure 40–50 mm in diameter versus 10–20 mm in smartphones. These larger drivers feature stiffer diaphragms and stronger suspension systems, requiring greater magnetic force to induce audible vibrations. Additionally, smart speakers incorporate more robust electromagnetic shielding in their audio amplifier circuitry, which attenuates external magnetic interference. Their thicker enclosures often include acoustic isolation materials that dampen internally generated sound before it reaches the microphone. Furthermore, multi-speaker configurations, such as the stereo pairs found in HomePod mini, create complex acoustic coupling paths that may reduce the efficiency of single-point magnetic injection. Laptops demonstrate

TABLE III
EXPERIMENT DEVICES, CATEGORY, SYSTEMS, AND RESULTS. WE EVALUATE MAGSTEALTH FROM INJECTION SUCCESS RATE, PESQ, AND SPL IN AN OFFICE ENVIRONMENT. VA: VOICE ASSISTANT, ISR: INJECTION SUCCESS RATE.

Num.	Category	Devices	Manufacturer	VA	ISR (%)	PESQ	SPL (dB)
1	Smartphone	Pixel 4	Google	X-Vector	96.5	3.83	33.5
2	Smartphone	Honor 50	Huawei	D-TDNN	96.1	4.04	32.8
3	Smartphone	Mate 20Pro	Huawei	ECAPA	96.4	3.86	36.1
4	Smartphone	Mate 30Pro	Huawei	Celia	97.4	4.02	36.3
5	Smartphone	Mate P40	Huawei	Celia	95.7	3.97	31.4
6	Smartphone	Nova 7	Huawei	Celia	95.4	4.04	33.9
7	Smartphone	Find x20Pro	Oppo	Breeno	95.2	3.89	33.5
8	Smartphone	Reno Pro	Oppo	Breeno	96.3	3.86	35.1
9	Smartphone	MI 5s Plus	Xiaomi	Xiaoai	97.5	4.05	35.6
10	Smartphone	MI 10	Xiaomi	Xiaoai	98.0	3.87	36.0
11	Smartphone	Redmi K50	Xiaomi	Xiaoai	97.7	4.00	32.2
12	Smartphone	Galaxy S10+	Samsung	Bixby	96.6	4.05	35.4
13	Smartphone	Galaxy S20+	Samsung	Bixby	96.8	4.01	35.2
14	Smartphone	iPhone 6s	Apple	Siri	97.0	3.76	31.3
15	Smartphone	iPhone 11	Apple	Siri	95.1	3.75	30.9
16	Smartphone	iPhone 12	Apple	Siri	96.3	3.84	33.9
17	Tablet	iPad	Apple	Siri	95.2	3.78	37.6
18	Tablet	iPad Pro	Apple	Siri	94.9	3.86	32.7
19	Laptop	Surface Go 2	Microsoft	Cortana	92.6	3.84	34.6
20	Laptop	ThinkPad T490	Lenovo	Cortana	93.3	3.72	31.7
21	Laptop	Xiaoxin 15	Lenovo	Cortana	92.3	3.78	36.0
22	Laptop	MacBook Pro	Apple	Siri	93.9	3.62	32.0
23	Speaker	HomePod mini	Apple	Siri	92.3	N/A	33.8
24	Speaker	MI Smart Speaker	Xiaomi	Xiaoai	92.0	N/A	34.4
25	Speaker	TmallGenie	AliGenie	TmallGenie	90.9	N/A	35.5

intermediate performance. This is likely due to their larger chassis volume, which increases the acoustic path length between the loudspeaker and microphone.

C. Robustness Analysis

1) *Impact of Position:* Given that the target devices may be located at various positions relative to MagStealth, it is crucial to assess how the position of the victim devices influences the effectiveness of the attack. This includes examining the impact of different attack distances and off-axis angles. In this experiment, we select the MI 5 s Plus, iPhone 12, and iPhone 6 s as target devices and compute the average results across all tests.

Orientation: The results presented in Fig. 15 reveal that MagStealth has a relatively omnidirectional attack range. Specifically, when the angle between the attack source and the target device is varied from -60° to 60° within a 30 mm distance, the injection success rate remains consistently above 94.7%. This demonstrates that MagStealth is highly adaptable to different device orientations, offering a wide attack angle that significantly enhances its practicality. The system's broad attack range makes it easy to deploy in real-world scenarios, where precise alignment with the target device may not always be possible.

Distance: The injection success rate remains at an average of 89.6% at a distance of 42 mm, with performance gradually declining as the distance increases. This is due to the natural decay of the energy in the emitted magnetic fields, which diminishes over greater distances, especially considering the limited power supply available in our compact attack prototype. Despite this reduction in effectiveness at longer ranges, the relatively short attack distance is still suitable for many practical applications. The compact design of MagStealth allows it to be discreetly hidden in inconspicuous locations, such as behind a table or within a charging socket. This makes it highly flexible and always ready to execute an attack whenever needed, without being easily detected.

2) *Impact of Ambient Noise:* Ambient noise can potentially affect the performance of MagStealth during a magnetic injection attack. However, since MagStealth generates acoustic signals internally within the device, it is expected to perform effectively even in noisy environments. To evaluate this, we conduct a series of attack experiments under varying levels of background noise with different SPL settings. In these experiments, the MI 5 s Plus, iPhone 12, and iPhone 6 s are selected as target devices. Two loudspeakers are positioned on either side of the device, emitting noise at a distance of 10 cm. The background

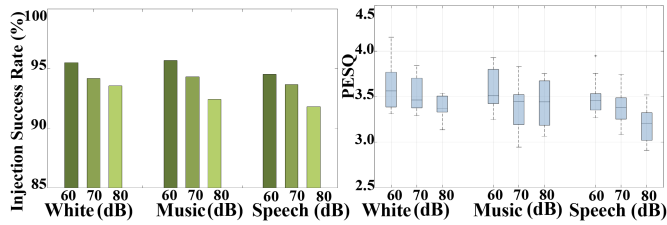


Fig. 14. Injection results under different types and SPLs of ambient noise.

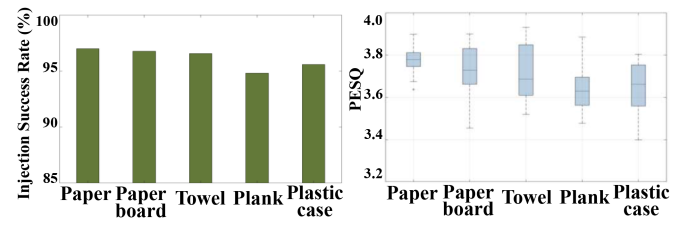


Fig. 16. Injection results under different types of occlusions.

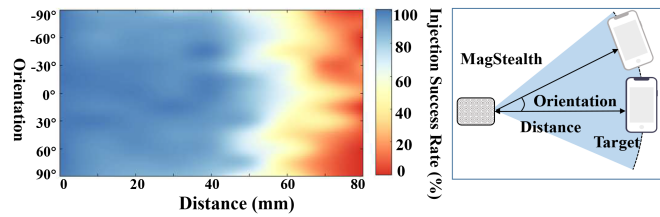


Fig. 15. Attacking performance under different positions.

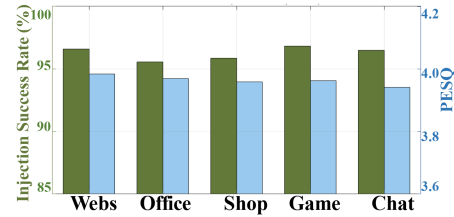


Fig. 17. Impact of running application.

noise sources include white noise, music, conversation, and traffic noise, representing common environmental sounds. The results, depicted in Fig. 14, indicate that MagStealth maintains strong performance under moderate noise conditions. At an interference level of 60 dB, the ISR remains nearly unchanged compared to quiet environments, demonstrating effective noise suppression inherent to the magnetic induction mechanism. Even when exposed to more intense noise levels, such as 80 dB, which is typically considered disruptive, MagStealth still maintains an impressive injection success rate above 93%, with a PESQ score of 3.57. These findings highlight the robust anti-noise capabilities of MagStealth, demonstrating its ability to reliably perform under challenging real-world conditions. This resilience is crucial for ensuring the effectiveness of MagStealth in dynamic, noisy environments where typical voice-command-based attacks may falter.

3) *Impact of Occlusion*: A critical factor in the stealth and practicality of magnetic injection attacks is the system's ability to operate effectively when the target device is not in direct line of sight. To assess this capability, we systematically evaluate the penetrability of MagStealth through a variety of common physical barriers by positioning different occluding materials between the attack setup and the target device. Six materials are selected to represent typical real-world obstructions that an attacker might use for concealment. These include paper with a thickness of 0.1 mm, cardboard of 3 mm, a cotton towel of 5 mm, a wooden plank of 8 mm, and a plastic case of 2.5 mm. Experimental results, shown in Fig. 16, indicate consistently high injection performance across all tested barriers. The average injection success rates under each occlusion are 97.0%, 96.8%, 96.6%, 94.8%, and 95.6%, respectively. Notably, even with the thickest obstruction, the 8 mm wooden plank, the success rate exhibits only a marginal decline of approximately 2%. Correspondingly, the PESQ metrics remain stable, with variations within 0.3 points, respectively. These findings confirm the robust penetrability of MagStealth through a range of non-conductive

and moderately conductive materials, which significantly extends its applicability in covert attack scenarios. The minimal degradation in performance when operating through occlusions highlights the advantage of magnetic field propagation, which is less affected by common physical barriers compared to acoustic or optical transmission, thereby enabling reliable and stealthy command injection even in obstructed environments.

4) *Impact of Running Application*: In addition to external factors, we also examine the performance of MagStealth when the target device is running various application programs, such as web browsers, office productivity software, video games, and instant messaging applications. In this experiment, we select the MI 5 s Plus, iPhone 12, and iPhone 6 s as the target devices. These devices are configured to run a diverse set of applications. The results, shown in Fig. 17, reveal that the performance of MagStealth remains consistently strong, regardless of the application running on the target device. The injection success rate and speech quality (as measured by PESQ) are stable, with only minimal variations across different types of applications. These findings suggest that the type of application does not significantly affect the effectiveness of the attack. This robustness underscores the versatility of MagStealth, demonstrating that its performance is not compromised by the concurrent operation of various applications. As a result, MagStealth is highly reliable for executing attacks in real-world scenarios, where devices are often multitasking and running multiple applications simultaneously.

5) *Impact of Injected Content*: Beyond evaluating overall performance on standard English voice commands, we further investigate how the semantic and linguistic characteristics of out-of-band modulated signals affect MagStealth's detection accuracy. To systematically assess this, we classify the voice data to be injected into three distinct categories: commands, speech, and dialogues. Additionally, to examine cross-linguistic robustness, we construct two new spoofed datasets comprising voice commands in Chinese and Spanish. As illustrated in

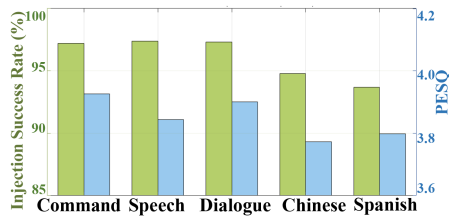


Fig. 18. Impact of injected content.

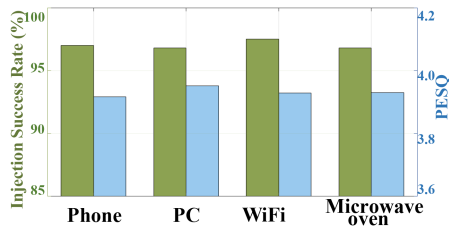
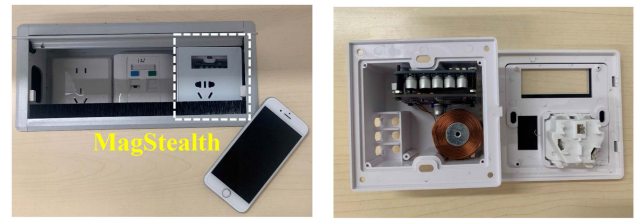


Fig. 19. Impact of electromagnetic leakage.

Fig. 18, MagStealth maintains a high average injection success rate above 93.7% across all content types and languages, demonstrating strong robustness regardless of semantic variation. We observe a modest decline in detection performance when the injected language is not English. This can likely be attributed to the fact that the adversarial signal generation module is pre-trained predominantly on English samples, thereby learning more discriminative feature representations for English inputs. Nevertheless, the system remains highly effective even under cross-lingual conditions, confirming its applicability in multi-lingual real-world environments.

6) *Impact of Electromagnetic Leakage:* In realistic environments, target devices are frequently situated near other operating electronics, which emit varying levels of electromagnetic interference (EMI). Such ambient EMI could potentially disrupt magnetic signal injection. To evaluate the resilience of MagStealth under these conditions, we conduct experiments in which the target device is placed in close proximity to several common electronic sources of EMI: an active smartphone, a desktop computer, a Wi-Fi router, and a microwave oven. The interfering electronic devices were positioned approximately 3 cm from the target device to emulate realistic electromagnetic leakage conditions. Typical operating transmission power levels were also considered during the experiments, including approximately 15–18 dBm for smartphones and around 18–20 dBm for Wi-Fi routers, while the microwave oven was operated under normal household usage conditions. As illustrated in Fig. 19, MagStealth maintains strong performance across all tested EMI scenarios. The injection success rate remains between 96.8% and 97.5%, while the PESQ varies only slightly, averaging around 3.9. These results demonstrate that the system is highly robust to typical levels of electromagnetic leakage encountered in everyday settings. This robustness can be attributed to the design of the magnetic injection signal, which operates in a frequency band and modulation scheme that are inherently resistant to common forms of broadband EMI. Furthermore, the direct



(a) Attack scenario.

(b) Inside the charging socket.

Fig. 20. MagStealth in a charging socket.

electromagnetic coupling between the attack transducer and the target device's internal loudspeaker provides a signal pathway that is largely isolated from airborne or radiated interference. Consequently, MagStealth proves capable of reliable operation even in electromagnetically crowded environments, affirming its practical applicability for stealth attacks in real-world scenarios.

To evaluate potential interference from internal device activity, we measure injection performance under idle conditions, CPU stress, active wireless transmission, and combined stress. The results show no significant variation, with standard deviation below 1.2 percent for injection success rate and 0.15 for PESQ. This robustness comes from the fact that magnetic injection directly couples to the passive voice coil in the analog domain. As a result, it is largely unaffected by internal computation and wireless activity. However, active noise cancellation mechanisms may introduce interference in some designs.

D. Real-World Experiment

While the preceding experiments were conducted under controlled laboratory conditions, it is essential to evaluate the practical applicability of MagStealth in more dynamic and realistic environments. To this end, we deploy a covert attack prototype integrated into a commercially available charging socket, as illustrated in Fig. 20. This modified socket houses the MagStealth hardware, allowing it to operate inconspicuously in everyday settings. The experiment is conducted in an active meeting room with ambient noise levels ranging from 30 dB to 70 dB, simulating a typical office or public space. The target devices, i.e., the MI 5 s Plus, MI 10, iPhone 6 s, and iPad Pro, are placed in close proximity to the modified charging socket, as would commonly occur during routine charging use. The attacker remotely activates the magnetic injection system via a Bluetooth connection, initiating a sequence of 20 injection attempts per device. The ISR and PESQ are recorded and averaged across all trials.

Results indicate that MagStealth achieves an average injection success rate of 96.25%, with a corresponding average PESQ of 3.96. These figures confirm that the system remains highly effective even in an acoustically varied and electromagnetically unstructured real-world environment. The majority of adversarial voice commands are successfully delivered and recognized by the target devices, demonstrating that magnetic field-based injection can reliably bypass both physical concealment and ambient interference. This experiment underscores the practical

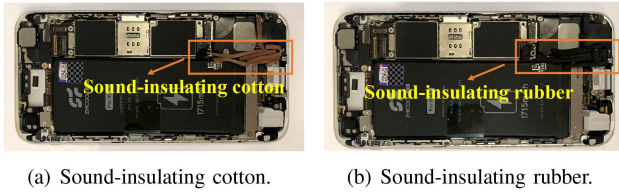


Fig. 21. The setup of sound blocking against MagStealth.

threat posed by MagStealth. By embedding the attack mechanism into a common charging accessory, an adversary can maintain persistent, stealthy access to nearby voice-enabled devices without raising suspicion. The results not only validate the technical feasibility of magnetic injection in realistic scenarios but also highlight the importance of considering such hardware-level attack vectors in the security design of voice-controlled systems.

IX. COUNTERMEASURE

Our study demonstrates that magnetic injection attacks pose a serious threat to a wide range of commercial electronic devices. To mitigate this threat, we propose a set of countermeasures from both hardware redesign and software-based detection perspectives, each targeting different attack vectors and deployment constraints.

Sound Blocking: The fundamental vulnerability exploited by magnetic injection stems from the physical proximity between the loudspeaker and microphone in typical audio system layouts. One direct hardware defense is to increase the physical separation between these components, or to insert sound-absorbing barriers [42] that attenuate direct acoustic coupling. However, such modifications often incur additional cost, size, and design complexity. To evaluate the effectiveness of material-based sound blocking, we conduct experiments (see Fig. 21) using two common insulating materials, i.e., sound-insulating cotton and rubber, wrapped around the built-in loudspeakers of test devices. After 30 magnetic injection attempts, the success rates drop to 87% for cotton and 70% for rubber, indicating that such materials can partially suppress internally induced audio leakage. While promising, material-based blocking alone is insufficient for complete protection, especially in compact devices where spacing is limited.

Electromagnetic Shielding: Beyond acoustic decoupling, electromagnetic shielding can be employed to protect sensitive audio components from external magnetic fields. A Faraday cage [43] represents a comprehensive shielding solution, but enclosing entire devices is often impractical for everyday use. A more feasible approach is to integrate localized shielding elements, such as ferrite beads [44], into critical circuit points (e.g., amplifier inputs or connector lines) to filter out high-frequency electromagnetic interference. Although these modifications add marginal cost, they can significantly raise the barrier against magnetic injection. To empirically assess shielding effectiveness, we conduct two experiments (Fig. 22): (1) placing an iPhone 6 s inside a professional Faraday cage (20 cm × 16 cm

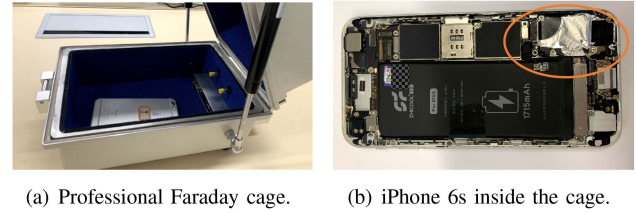


Fig. 22. The setup of electromagnetic shielding against MagStealth.

× 16 cm), and (2) wrapping only its loudspeaker tightly with tin foil. Attacks launched from various distances and angles against the Faraday-caged device all fail, confirming that a properly grounded enclosure can fully block external magnetic signals. Note, however, that such shielding also disrupts legitimate wireless communication (Wi-Fi, cellular, etc.). In contrast, attacks against the tin-foil-wrapped device remain successful, demonstrating that thin, ungrounded metallic layers provide negligible magnetic attenuation.

Liveness Detection: A complementary software-based defense is voice liveness detection, which discriminates between genuine human speech and artificially injected signals by exploiting physiological and acoustic cues. To evaluate this approach against MagStealth, we collect a dataset of 1000 injected samples via magnetic attack and 1000 genuine human speech samples. Following prior work that highlights spectral and dynamic differences between live and injected voices, we extract a feature set comprising Mel-frequency cepstral coefficients (MFCCs), formant trajectories, and fundamental frequency contours. These features are used to train a Transformer-based classifier.

The liveness detection model is implemented using a Transformer-based classifier consisting of four encoder layers, each equipped with eight-head self-attention and feed-forward layers. The input feature representation includes MFCC coefficients, formant trajectories, and fundamental frequency contours extracted from the audio signals. The model is trained using the Adam optimizer with a learning rate of $1e-4$, a batch size of 32, and 50 training epochs. The dataset is divided into 80% training samples and 20% testing samples.

In testing, the classifier achieves an average detection accuracy of 84.6%, indicating that learned acoustic-physiological patterns can effectively flag many magnetic injections. However, when the training set contains only genuine speech, accuracy drops to 47.8%, essentially random guessing, highlighting the model's dependence on exposure to attack signatures. This reliance poses a practical challenge: it is infeasible to collect exhaustive training data covering all possible attack variants, devices, and environmental conditions. It is important to note that the limitation of current liveness detection approaches is primarily practical rather than fundamental. Most detection models rely on supervised learning and require representative attack samples during training [45]. Since collecting comprehensive datasets covering all possible attack variants is challenging, existing detectors may struggle to generalize to previously unseen magnetic injection attacks. The current evaluation focuses

on specific device configurations used in our experiments and does not claim full cross-device generalization. Evaluating the robustness of liveness detection across a wider range of hardware platforms remains an important direction for future work.

X. DISCUSSION

A. Limitations

Physical and System-Level Limitations: The effectiveness of MagStealth is intrinsically tied to the rate of change of magnetic flux induced in the target device. Consequently, any movement or vibration of the device during an attack can alter the magnetic coupling and potentially degrade the injection performance. For optimal results, the attack is most effective against stationary targets. Additionally, the operational range of MagStealth is currently limited to approximately 8 cm, necessitating close physical proximity to the target. However, the compact form factor of the prototype allows it to be discreetly embedded within common objects such as power sockets, making it feasible to deploy in public or semi-public settings including meeting rooms, cafes, and lounges. When integrated into a modified charging socket, an attacker can opportunistically activate voice assistants on devices being charged. While MagStealth can inject commands into locked smartphones, execution of those commands often requires user interaction; for example, iPhones typically do not perform sensitive actions without unlocking. However, some Android models (e.g., certain OPPO phones) may still execute limited commands such as enabling Bluetooth or Wi-Fi, presenting a tangible security risk even when the device is locked. From a physical perspective, magnetic fields can penetrate most materials, with penetration depth determined by the permeability of the medium and the frequency of the excitation current [46]. Nevertheless, certain configurations can significantly attenuate the attack. For instance, if the target device is surrounded by conductive metal enclosures, time-varying magnetic fields from MagStealth may induce eddy currents within the metal. These currents generate opposing magnetic fields that partially cancel the injected field, reducing the effective signal strength at the target. Increasing the thickness or number of metal barriers amplifies this shielding effect. Furthermore, environmental factors such as elevated temperature [47] or exposure to corrosive liquids [48] can degrade the magnetic properties of the attack hardware, effectively neutralizing the threat.

Responsible Disclosure and Prevention Guidance: We have responsibly disclosed the MagStealth vulnerability to major manufacturers including Apple, Google, Samsung, Huawei, and Xiaomi prior to publication, providing technical details and proof-of-concept demonstrations. Based on our findings, we recommend several design-phase protections. Hardware-level isolation through increased separation between loudspeaker and microphone or acoustic baffles can attenuate internally coupled audio. Magnetic shielding with mu-metal or ferrite around voice coils and amplifier stages can block external magnetic fields, as properly grounded enclosures completely prevent injection. At the system level, stricter secure boot and trusted execution policies should ensure that voice assistant services cannot perform sensitive operations when the device is locked. Enhancing

liveness detection models with simulated magnetic injection data can improve robustness, especially in unseen attack scenarios. And we will share our datasets with manufacturers under non-disclosure agreements to support defense development. We emphasize that this work is intended to advance understanding of emerging hardware-level threats and to promote effective countermeasures, and we strongly discourage any malicious use of the proposed technique.

Operating System and Lock-State Considerations: The impact of MagStealth injection depends on the device's operating system and lock-screen policies. Our experiments across iOS, Android, and HarmonyOS revealed significant variation in locked-state voice assistant accessibility. On iOS, locked devices permit limited Siri commands such as time queries, alarm setting, and music control, while blocking personal data access and transactions. This offers partial protection but still enables malicious actions like enabling Bluetooth. Android behavior varies by manufacturer. Google Pixel devices enforce iOS-like restrictions, while some OPPO and Xiaomi models allow broader locked-state execution including Wi-Fi and Bluetooth enabling, app opening, and calling contacts, creating a heightened security risk. Smart speakers require no authentication for any voice command, making them the most vulnerable category despite lower injection success rates. Laptops generally follow their mobile OS counterparts' lock-screen policies. Attackers can probe device state with innocuous queries before escalating to sensitive operations or chain commands to gradually compromise security.

B. Future Work

Potential Improvements: The current MagStealth prototype is powered by a 12 V, 1 A supply, which prioritizes portability but constrains the effective attack range. Increasing the power supply capacity and employing multiple synchronized electromagnets could extend the operating distance from centimeters to potentially meter-scale ranges. Moreover, the cylindrical electromagnet used in the present design retains a portion of its magnetic field internally rather than radiating it outward. Through electromagnetic simulation tools such as COMSOL and Ansys [49], the coil geometry could be optimized to enhance external field distribution and coupling efficiency. Another promising direction is to modulate the injection signal into the ultrasonic frequency band (e.g., around 20 kHz), leveraging techniques similar to DolphinAttack [7] to exploit nonlinearities in microphone responses, thereby further masking the attack from human hearing.

Other Potential Targets for Magnetic Attacks: This work demonstrates magnetic injection via the loudspeaker's electromechanical pathway. The same underlying principle can be extended to other magnetic-sensitive components. For example, dynamic microphones [50] operate on the inverse mechanism of loudspeakers: acoustic vibrations move a coil within a magnetic field, generating an electrical signal. Applying a controlled external magnetic field could therefore inject arbitrary audio signals directly into the microphone input, bypassing physical acoustics. Transformers [51], which rely on electromagnetic

TABLE IV
COMPARISON OF MAGSTEALTH WITH REPRESENTATIVE ATTACK METHODS

Attack Method	Attack Medium	Penetrability	Max Range	Hardware Cost	Bypasses Speaker ID	Acoustic Leakage
MagStealth	Magnetic field	High (through materials)	8 cm	Low (<\$50)	Yes (adversarial)	None (internal)
DolphinAttack [7]	Ultrasound	Low (blocked by materials)	1.5 m	Low	No	None
Backdoor [6]	Ultrasound	Low	1.2 m	Low	No	None
Light Commands [16]	Laser	Very Low (line-of-sight)	110 m	High (>\$500)	No	None
CommanderSong [10]	Audible audio	Low (airborne)	5 m	Very Low	Limited	Audible
UltraAdv [8]	Ultrasound	Low	2 m	Medium	Limited	None
LaserAdv [17]	Laser	Very Low	50 m	High	Yes (adversarial)	None

induction to transfer energy between circuits, represent another potential target. By exposing a transformer to a maliciously varying external magnetic field, an attacker could destabilize its output voltage, leading to malfunctions in power supplies or disruptions in communication networks that employ transformer-based isolation. Investigating these vectors would broaden the understanding of magnetic interference threats across a wider spectrum of electronic systems.

XI. RELATED WORK

Injection Attacks on Voice Assistants: A growing body of research has demonstrated that embedded microphones in voice-controlled systems can serve as unintended backdoors, enabling attackers to inject malicious commands. Existing injection techniques can be broadly classified into audible and inaudible attacks. (1) *Audible Attacks.* Early studies identified vulnerabilities in voice-controlled systems under mimicry attacks [52] and replay attacks [53]. More recent work has shifted towards embedding adversarial perturbations into audible media such as music [9], [10], [11], [54], [55]. These perturbations are designed to be recognized by voice assistants while remaining unintelligible to human listeners. Other audible approaches exploit phonetic similarities to trigger misclassification of wake-words [56], [57]. In contrast to these externally audible methods, MagStealth operates in a more covert manner by confining the injected audio entirely within the interior of the target device, leaving no audible trace in the surrounding environment. (2) *Inaudible Attacks.* Another line of work exploits the nonlinear response of microphones to ultrasonic frequencies. Seminal studies such as Backdoor [6] and DolphinAttack [7] demonstrated that modulated ultrasonic signals can carry inaudible commands that are demodulated by the microphone's hardware. Subsequent work extends the effective range of ultrasound-based injection [58] and explores alternative propagation media such as solid materials [59] or unconventional sound sources [60], [61]. Compared with existing command injection attacks such as ultrasound-based attacks and laser-based attacks, as shown in Table IV, MagStealth provides several distinct advantages. Ultrasound attacks rely on microphone nonlinearity and often suffer from strong signal attenuation and limited penetration through obstacles. Laser-based attacks require precise line-of-sight alignment and specialized optical equipment, which significantly increases

deployment cost and complexity. In contrast, MagStealth leverages magnetic field penetration and internal loudspeaker coupling, enabling obstacle penetration, omnidirectional coverage, and lower hardware cost, making it more practical for stealthy command injection scenarios.

Research on Magnetic Security: Significant research efforts have been devoted to exploring magnetic fields for both offensive and defensive security applications. On the offensive side, magnetic covert channels have been proposed for data exfiltration from air-gapped systems by modulating CPU load into magnetic emissions [62], [63], [64]. Magnetic side-channel attacks [65], [66] have also been demonstrated against hard drives, where smartphone-based magnetic sensors can fingerprint drive activity [67]. Furthermore, ambient magnetic fluctuations caused by indoor structures have been exploited for indoor localization, raising privacy concerns [68]. On the defensive side, magnetic sensing has been utilized for security applications such as detecting loudspeaker emissions to counter voice impersonation attacks [52] and enabling gesture-based authentication resistant to shoulder-surfing [69], [70]. While prior work has largely focused on magnetic sensing and covert channels, this paper focuses specifically on the threat of active magnetic injection. MagStealth represents a novel offensive application of magnetic fields, introducing a practical hardware prototype for stealthy command injection via electromagnetic coupling to internal audio components.

XII. CONCLUSION

This work reveals a critical hardware-level vulnerability in smart voice devices: their built-in loudspeakers can be externally manipulated to execute stealthy command injection attacks. We present MagStealth, a novel magnetic field-based attack that induces malicious audio commands via the electromagnetic-acoustic coupling of internal loudspeakers, bypassing common defenses such as noise, physical obstructions, and speaker verification systems. Through comprehensive simulation and experimental studies, we establish a theoretical model that quantifies the influence of external magnetic fields on loudspeaker acoustics, providing foundational insights for magnetic injection techniques. To enable covert and practical execution, we develop a compact hardware prototype integrated with a diffusion-based adversarial signal generator, allowing the system to be concealed within everyday objects such as charging sockets. Extensive evaluation across 25 commercial devices demonstrates

that MagStealth achieves an average injection success rate of over 95% while maintaining high audio fidelity and operational stealth under diverse real-world conditions.

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Tiantian Liu received the bachelor's degree from Zhejiang University, in 2020, and the PhD degree from Zhejiang University, in 2025. She is an assistant professor with the School of Informatics, Xiamen University. Her research interests include cyber-physical systems (CPS) security, speech security, AI security, and mobile computing. Her work received the Best Paper Candidate Award from ACM SenSys 2021 and SIGMOBILE Research Highlight. She was also named the EECS Rising Star 2024 by MIT.



Peng Huang received the BS degree in Information Engineering from Zhejiang University, China, 2020. He is currently working towards the PhD degree with the School of Cyber Science and Technology, Zhejiang University, supervised by Prof. Zhongjie Ba. His research interests include IoT Security, AI Security, and Privacy-preserving Technologies. He has authored or co-authored papers published in top conferences and journals, such as NDSS, IEEE S&P, and ACM MobiSys.



Feng Lin (Senior Member, IEEE) received the PhD degree from the Department of Electrical and Computer Engineering, Tennessee Technological University, in 2015. He is currently a professor with the School of Cyber Science and Technology, College of Computer Science and Technology, Zhejiang University, China. He was an assistant professor with the University of Colorado Denver, a research scientist with the State University of New York (SUNY) at Buffalo, and an engineer with Alcatel-Lucent (currently, Nokia). His current research interests include mobile sensing, wireless sensing, Internet of Things security, biometrics, and AI security. He was a recipient of the ACM SIGSAC China Rising Star Award, the Best Paper Awards from ACM MobiSys'20, IEEE Globecom'19, IEEE BHI'17, the Best Demo Award from ACM HotMobile'18, and the Best Paper Award Nomination from SenSys'21 and INFOCOM'21. He serves as an editor for *IEEE Network*.



Tao Ni received the bachelor's degree from Shanghai Jiao Tong University, in 2018, the master's degree from Australian National University, in 2020, and the PhD degree from the Department of Computer Science, City University of Hong Kong. He is an assistant professor with the Computer, Electrical and Mathematical Sciences and Engineering (CEMSE) Division, King Abdullah University of Science and Technology (KAUST). He was a postdoctoral research fellow with the Department of Computer Science, City University of Hong Kong. His research interests include cyber-physical systems (CPS) security, AI security, and mobile computing. His work received the 2024 Cybersecurity Best Practical Paper Award and was also named an ACM MobiSys Rising Star (2024) and an AIOtsys Rising Star (2025).



Liang Xiao (Fellow, IEEE) received the BS degree in communication engineering from the Nanjing University of Posts and Telecommunications, China, in 2000, the MS degree in electrical engineering from Tsinghua University, China, in 2003, and the PhD degree in electrical engineering from Rutgers University, New Jersey, in 2009. She was a visiting professor with Princeton University, Virginia Tech, and the University of Maryland, College Park. She is currently a professor with the Department of Information and Communication Engineering, Xiamen University, Xiamen, China. She was a recipient of the Best Paper Award for 2016 INFOCOM Big Security WS and 2017 ICC. She has served as an associate editor for *IEEE Transactions on Information Forensics and Security* and a guest editor for *IEEE Journal of Selected Topics in Signal Processing*.